ELEC-271, Digital Systems Fall 2026
Midterm Formula Sheet

Section 1 Binary Numbers

Positional number representation

convert from base k to decimal value:

$$Value = \sum_{i=-m}^{n-1} b_i . k^i$$

n is the number of digits before decimal point.
m is the number of digits after decimal point.
b_i is digit at position i.
k The base of the original number (e.g., k=2 for binary, k=16 for hexadecimal).

Binary to decimal conversion

$$B = b_{n-1}b_{n-2}\cdots b_1 b_0 \qquad V(B) = b_{n-1}\times 2^{n-1} + b_{n-2}\times 2^{n-2} + \cdots + b_1 \times 2^1 + b_0 \times 2^0 = \sum_{i=0}^{n-1} b_i \times 2^i$$

B represents number in binary format, $V(B)$ is number in decimal format

Conversion table

Conversion Table

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Section 2 Logic Circuits

Boolean Algebra

1a.	$0 \cdot 0 = 0$	1b.	$1 + 1 = 1$
2a.	$1 \cdot 1 = 1$	2b.	$0 + 0 = 0$
3a.	$0 \cdot 1 = 1 \cdot 0 = 0$	3b.	$1 + 0 = 0 + 1 = 1$
4a.	If $x = 0$ then $\overline{x} = 1$	4b.	If $x = 1$ then $\overline{x} = 0$

5a.	$x \cdot 0 = 0$	5b.	$x + 1 = 1$
6a.	$x \cdot 1 = x$	6b.	$x + 0 = x$
7a.	$x \cdot x = x$	7b.	$x + x = x$
8a.	$x \cdot \overline{x} = 0$	8b.	$x + \overline{x} = 1$
9.	$\overline{\overline{x}} = x$		

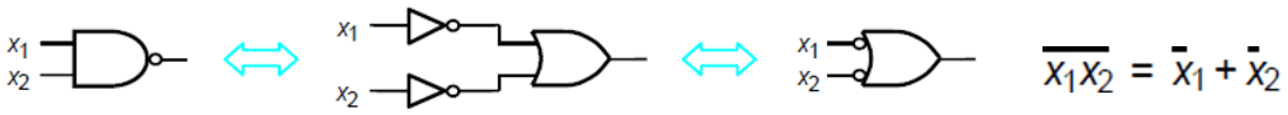
10a.	$x \cdot y = y \cdot x$	10b.	$x + y = y + x$	*Commutative*
11a.	$x \cdot (y \cdot z) = (x \cdot y) \cdot z$	11b.	$x + (y + z) = (x + y) + z$	*Associative*
12a.	$x \cdot (y + z) = x \cdot y + x \cdot z$	12b.	$x + y \cdot z = (x + y) \cdot (x + z)$	*Distributive*
13a.	$x + x \cdot y = x$	13b.	$x \cdot (x + y) = x$	*Absorption*
14a.	$x \cdot y + x \cdot \overline{y} = x$	14b.	$(x + y) \cdot (x + \overline{y}) = x$	*Combining*
15a.	$\overline{x \cdot y} = \overline{x} + \overline{y}$	15b.	$\overline{x + y} = \overline{x} \cdot \overline{y}$	*DeMorgan's Thm*
16a.	$x + \overline{x} \cdot y = x + y$	16b.	$x \cdot (\overline{x} + y) = x \cdot y$	
17a.	$x \cdot y + y \cdot z + \overline{x} \cdot z = x \cdot y + \overline{x} \cdot z$	17b.	$(x + y) \cdot (y + z) \cdot (\overline{x} + z) = (x + y) \cdot (\overline{x} + z)$	*Consensus*

Minterm and Maxterm table (3 input variable case)

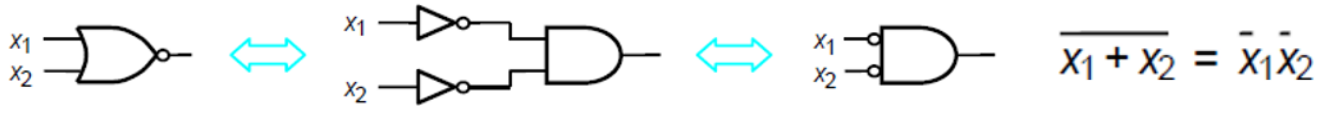
Row number	x_1	x_2	x_3	Minterm	Maxterm
0	0	0	0	$m_0 = \overline{x}_1\overline{x}_2\overline{x}_3$	$M_0 = x_1 + x_2 + x_3$
1	0	0	1	$m_1 = \overline{x}_1\overline{x}_2x_3$	$M_1 = x_1 + x_2 + \overline{x}_3$
2	0	1	0	$m_2 = \overline{x}_1x_2\overline{x}_3$	$M_2 = x_1 + \overline{x}_2 + x_3$
3	0	1	1	$m_3 = \overline{x}_1x_2x_3$	$M_3 = x_1 + \overline{x}_2 + \overline{x}_3$
4	1	0	0	$m_4 = x_1\overline{x}_2\overline{x}_3$	$M_4 = \overline{x}_1 + x_2 + x_3$
5	1	0	1	$m_5 = x_1\overline{x}_2x_3$	$M_5 = \overline{x}_1 + x_2 + \overline{x}_3$
6	1	1	0	$m_6 = x_1x_2\overline{x}_3$	$M_6 = \overline{x}_1 + \overline{x}_2 + x_3$
7	1	1	1	$m_7 = x_1x_2x_3$	$M_7 = \overline{x}_1 + \overline{x}_2 + \overline{x}_3$

$\overline{m}_i = M_i$

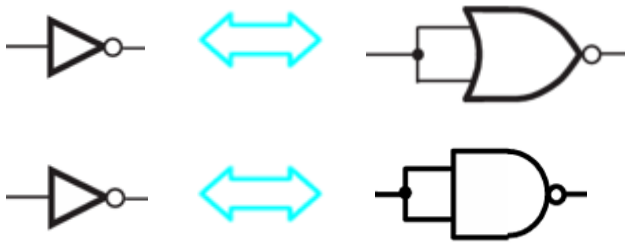
NAND and NOR gate equivalence



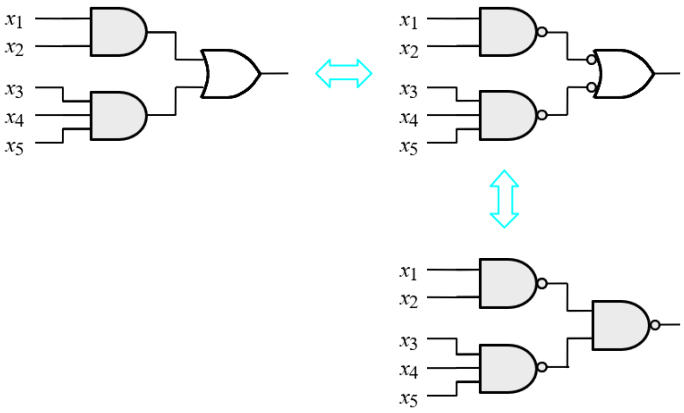
NAND gate



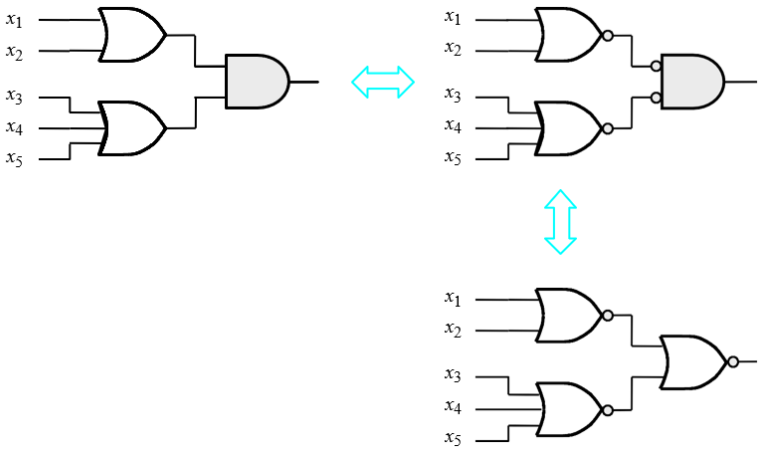
NOR gate



SOP to NAND-NAND network

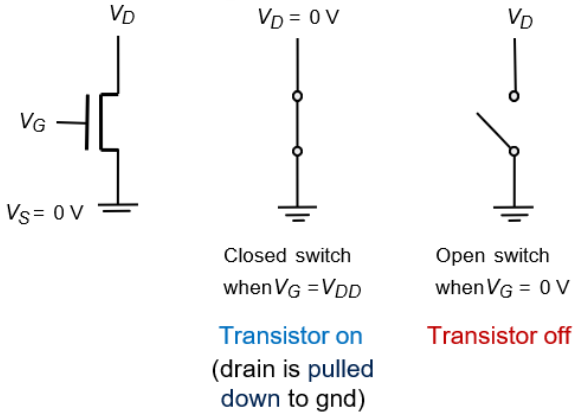


POS to NOR-NOR network



Section 3: Implementation Technology

NMOS transistor



PMOS transistor

